Langchain Academy

Foundation - Introduction to Langchain - Python

Foundational model → Thinking brain

Customize the model by tuning these parameters.

> temperature → to control the randomness

> max_tokens → how long the response will be

> timeout → how long to wait for the response

> max_retries → number of times to retry if the request fails.

Streaming → to show the output tokens as and when it gets generated.

System Prompt → asks the model to focus and stay within the designated chat.

OR

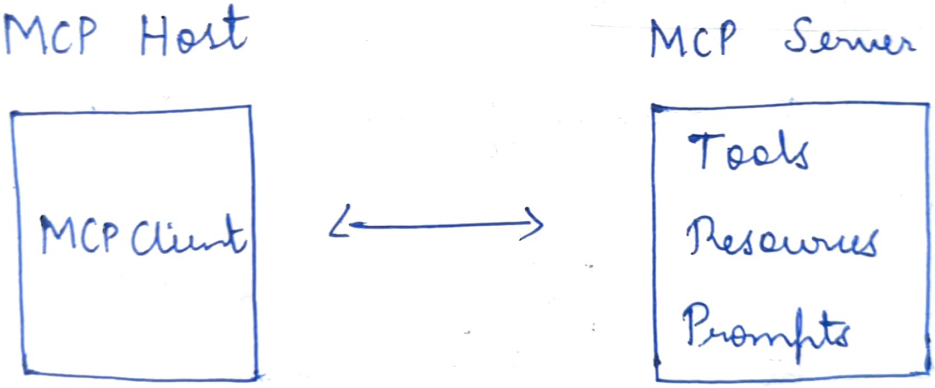
You can do the structured output schema

Short - Term Memory

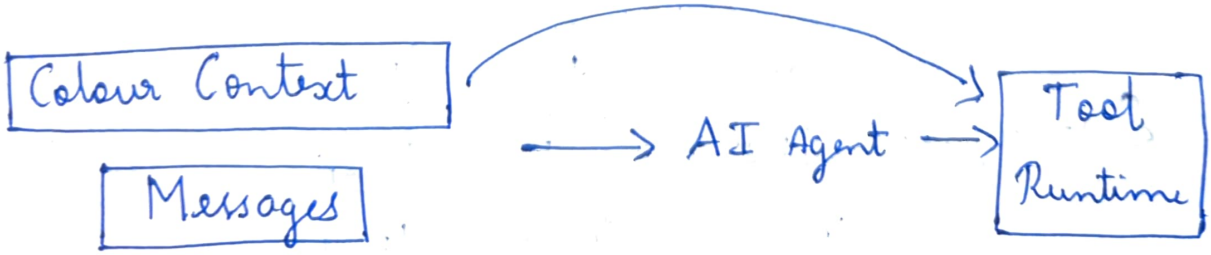
In Memory Saver module of Langchain will group all the conversations of a particular state into a single thread id and adds it to the agent.

MCP

Open protocol how LLM models connect with each other and other external tools. Like a USB-C port.



Context and State



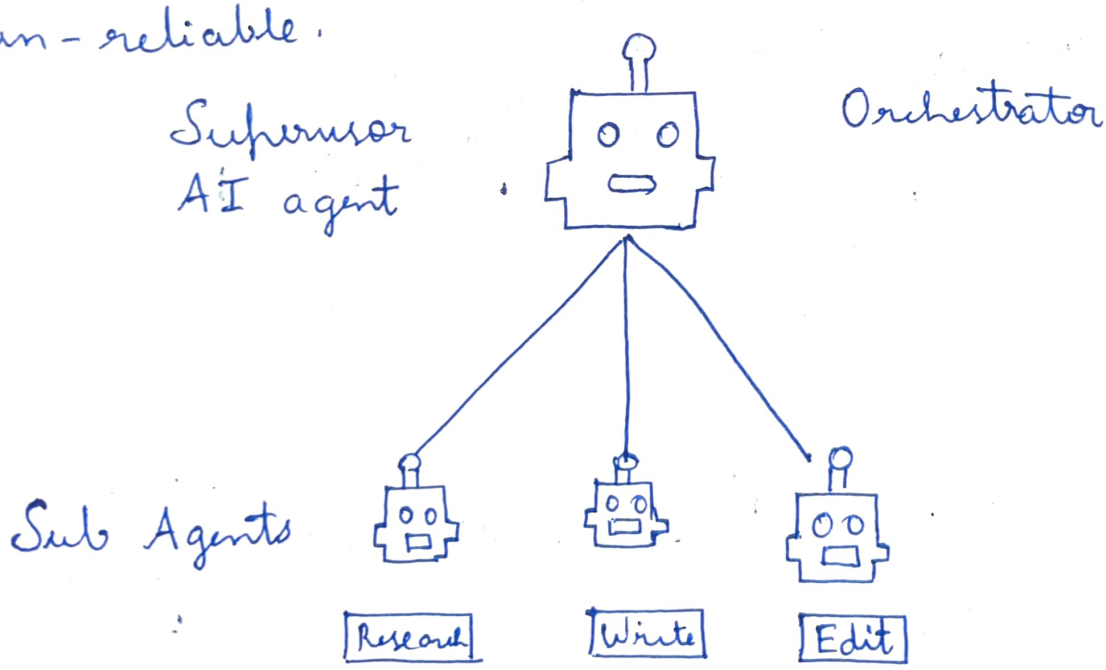
Instead of overloading our model with too much context window → we pass a tool call to get the context.

We can add <u>Custom State</u> for the AI to keep a track on what user wants. (save user's prep)

> We create a tool to pass the state → to avoid overloading of AI with context window.

<u>Multi Agent Systems</u>

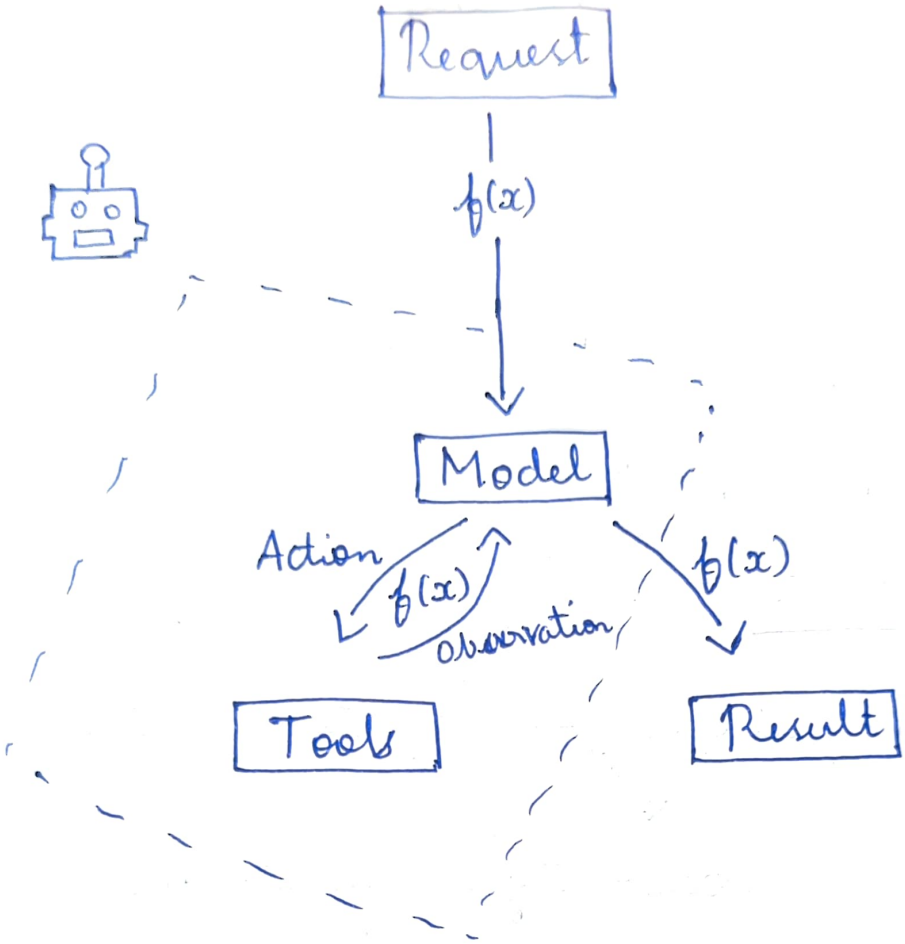
> Single Agent handling all the tasks is un-reliable.



Through <u>Langchain</u> you can add <u>sub agents</u> as <u>tools</u>

Production Ready Agent

Middleware



Function that we can insert within these loops that allows us to customize and control our agent execution

Managing Long Conversations

As the messages/conversations ↑ agents response time ↓, performance ↓.

So we can summarise all the conversations and delete few as a middleware function to maintain long conversations.

Long Conversations $\xrightarrow{\text{middleware}}$ summarize OR Delete / trim

Custom middleware funtions → provides granular level of deleting messages.

Human in the Loop

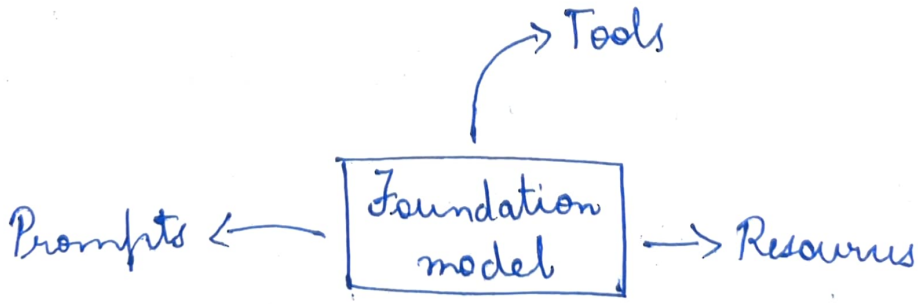
→ Approving sensitive actions

→ Debug agents

Dynamic Agents

Custom middleware → agents runtime
↓
node-style architecture

model instance → model request
↓



Through decorator style of code:

1.) If info is internal → sql database, web search

2.) " " external → web search

Wrap-Style Middleware	Node-style middleware	Tool Calls
Model request	State + Runtime	Tool runtime
Adjust an agent's tools, prompt, model while tuning	Adjust an agent's state while running	Agent adjusts its own state or retrieves runtime context
eg: Giving an agent access to tools if the user is internal	eg: Trimming messages from long conversations	eg: Updating the state with the username when provided to the agent